Video 1. Basic pumping action of a DEA diaphragm.
A circular DEA diaphragm (10 mm diameter), clamped around its edge, is driven by a 1 Hz sinusoidal voltage ranging from 0 to 2 kV (1 kV amplitude, 1 kV offset). At 0 V the diaphragm lies flat, and as the voltage increases, the membrane expands and bulges upward into a dome shape.

Video 2. Fluid flow produced by the DEA pump and Tesla valve system.
Split-screen video of water, visualized with red dye, being pumped from the inlet reservoir to the outlet reservoir. The right panel shows the entire integrated pump, and the left panel shows a magnified view of the outlet reservoir. The diaphragm is driven by a sinusoidal voltage ranging from 0 to 2 kV (1 kV amplitude, 1 kV offset). The pulsating flow is particularly evident at 5 Hz.